

API Plan: Decision-Support & Comparison Engine

Frontend pages: React displays the landing page, tutorial, registration, login, dashboard, decision editor, and results page. The Express routes below return data to React rather than displaying pages or redirecting the browser.

Account Routes

API Endpoint = POST => “/api/auth/register” => Create Account

- The user sends an email and password. The backend stores a password hash.
- A successful response returns 201 with the new user's ID and email.
- An invalid email, missing password, or email already in use returns 400 with an explanation.

API Endpoint = POST => “/api/auth/login” => Log In

- The user sends an email and password.
- A successful response returns 200 with the user's ID and email and establishes their login session.
- Incorrect credentials return 401.

API Endpoint = GET => “/api/auth/me” => Check Current User

- A successful response returns 200 with the signed-in user's ID and email. React can use this when the app opens or refreshes.
- If no one is signed in, the response returns 401.

API Endpoint = POST => “/api/auth/logout” => Log Out

- The backend ends the login session.
- A successful response returns 200 with a logout confirmation.

Decision Routes

API Endpoint = POST => “/api/decisions” => Save New Decision

- Requires login. Receives the decision's title, optional description, criteria and weights, options and scores, and optional chosen option and explanation.
- The backend identifies the owner from the signed-in user and calculates the rankings.
- A successful response returns 201 with the new decision's ID, saved decision, and rankings.
- Invalid decision data returns 400; no login returns 401.

API Endpoint = GET => “/api/decisions” => List Saved Decisions

- Requires login. Returns 200 with the signed-in user’s decisions.
- If they have none, returns 200 with an empty list.
- No login returns 401.

API Endpoint = GET => “/api/decisions/:decisionId” => View One Saved Decision

- Requires login. Returns 200 with that user’s decision and its calculated rankings.
- No login returns 401. If no accessible decision matches the ID, returns 404.

API Endpoint = PATCH => “/api/decisions/:decisionId” => Update Saved Decision

- Requires login. Updates the submitted parts of an existing decision, such as weights, options, scores, or the chosen option and explanation.
- A successful response returns 200 with the updated decision and recalculated rankings.
- Invalid data returns 400; no login returns 401; no accessible decision matches the ID returns 404.

API Endpoint = DELETE => “/api/decisions/:decisionId” => Delete Saved Decision

- Requires login. Deletes one of the signed-in user’s saved decisions.
- A successful response returns 200 with “Decision deleted successfully.”
- No login returns 401; no accessible decision matches the ID returns 404.

Guest Comparison Route

API Endpoint = POST => “/api/evaluations” => Calculate Results Without Saving

- Does not require login. Receives criteria with weights and options with scores.
- A successful response returns 200 with calculated totals and rankings. It does not save the decision to MongoDB.
- Missing scores or values outside the 1–5 range return 400 with an explanation.